Course Project Report

Smart Sparse Retrieval: Learning-Enhanced Information Retrieval using LLM-Based Query Expansion

Submitted By

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C E R T I F I C A T E

This is to certify that the Course project Work Report entitled **"Smart Sparse Retrieval: Learning-Enhanced Information Retrieval using LLM-Based Query Expansion "** is submitted by the group mentioned below -

Details of Project Group

Name of the Student	Register No.	Signature with Date
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this report is a record of the work carried out by them as part of the course **Information Retrieval (IT367)** during the semester **Jan-Apr 2026**. It is accepted as the Course Project Report submission in the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence.**

(Name and Signature of Course Instructor)
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D E C L A R A T I O N

We hereby declare that the project report entitled **"Smart Sparse Retrieval: Learning-Enhanced Information Retrieval using LLM-Based Query Expansion "** submitted by us for the course **Information Retrieval (IT367)** during the semester **Jan-Apr 2026**, as part of the partial course requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence at NITK Surathkal is our original work. We declare that the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles elsewhere.

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Date: April 15, 2026

Smart Sparse Retrieval: Learning-Enhanced Information Retrieval using LLM-Based Query Expansion

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***Abstract*— Sparse retrieval systems such as BM25 remain foundational to modern information retrieval pipelines owing to their efficiency and interpretability, yet they suffer from vocabulary mismatch when user queries fail to share exact lexical overlap with relevant documents. Large Language Models (LLMs) have recently demonstrated strong potential for mitigating this issue via zero-shot query expansion—generating hypothetical answer passages that enrich the query with semantically relevant vocabulary. In this work, we replicate and systematically extend the Exp4Fuse framework (Liu and Zhang, 2025), which fuses BM25 ranked lists derived from the original query and an LLM-augmented query through a modified Reciprocal Rank Fusion (RRF) scoring scheme. Building on the confirmed baseline parity on the TREC DL19 passage retrieval task, we investigate three original research directions: (i) a reduced hyperparameter grid search over the query repetition factor λ, the RRF constant *k*, and fusion weights; (ii) an ablation study comparing four distinct fusion methods—modified RRF, standard RRF, linear score combination, and CombSUM—across nDCG@10, MAP, and Recall@1000; and (iii) a controlled study on hypothesis truncation examining whether limiting the LLM-generated text to 64 or 96 tokens affects retrieval quality relative to the full hypothesis. Our experiments reveal that a smaller λ of 4 combined with *k*=30 yields the best nDCG@10 of 0.5535, that score-normalized fusion methods (CombSUM and linear combination) outperform rank-only methods on nDCG@10 by a margin of +0.020, and that hypothesis truncation at the tested lengths leaves retrieval metrics entirely unaffected—suggesting that BM25 is robust to the length of the appended pseudo-document beyond a certain threshold.**

***Keywords:* information retrieval, sparse retrieval, BM25, query expansion, large language models, reciprocal rank fusion, TREC DL19**

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I. Introduction

The information retrieval (IR) problem refers to locating pertinent documents in a large collection based on a query issued by a user. Sparse lexical IR models such as BM25 (Robertson and Zaragoza, 2009) have been the workhorses of initial-stage IR for years due to their speed, excellent empirical baselines, and term weighting interpretability. While they continue to prove valuable, however, sparse models suffer from the problem of vocabulary mismatch: if the terms in the user query are not present in the target document, then BM25 fails to retrieve the document, regardless of its semantic relevance (Zhao et al., 2021).

To mitigate the issue of lexical gaps, one may resort to dense retrievers, which exploit pre-trained transformer encoders to represent both queries and documents with vectorial embeddings that can match semantically similar items, even if queries are paraphrased versions of their corresponding documents (Karpukhin et al., 2020). However, this approach comes at the price of a computationally expensive inference process, expensive fine-tuning using labeled relevance annotations, and sensitivity to the retrieval domain. Hence, the use of sparse and dense retrievers in multi-step frameworks where sparse retrievers such as BM25 generate candidates and dense retrievers perform neural ranking.

A different and cost-effective way is called *query expansion* (QE). In QE, relevant terms are added to the original query of the user before sparse retrieval is performed. Traditional approaches for QE rely on terms extracted through pseudo-relevance feedback or thesaurus lookup (Carpineto and Romano, 2012). However, such methods face limitations based on the quality of the original retrieval and lack of knowledge coverage. With the introduction of large pre-trained language models capable of following instructions, a novel direction arises in the form of zero-shot QE. It consists of having an LLM generate a passage which would answer the query question in a hypothetically correct way, after which the passage is appended to the original query for retrieval. This technique was shown to significantly boost the performance of sparse and dense retrieval on benchmarks via methods like HyDE (Gao et al., 2023) and Query2Doc (Wang et al., 2023).

Exp4Fuse (Liu and Zhang, 2025), which was recently introduced as a solution to this problem, takes this approach further by decoupling the augmentation from the retrieval stage altogether. Instead of directly providing the concatenated query as input to the BM25 retriever, which may introduce noise due to hallucinations or domain mismatches from the LLM, Exp4Fuse performs two independent runs of the BM25 retrieval stage on both the base query and the expanded query produced by the LLM. These two rankings are then fused using an enhanced RRF operator, which rewards documents present in both rankings.

In this work, we (i) reproduce the baseline experiment Exp4Fuse in TREC DL19 passage retrieval task for consistency, and (ii) explore three different novel directions. We perform an exhaustive hyperparameter search on λ, the query repetition factor, *k*, the RRF coefficient, and the weighting factors on the fusion process for each routing path; we evaluate four different fusion methods, namely, modified RRF, original RRF, linear score interpolation, and CombSUM; and we investigate whether restricting the hypothesis generated by the LLM into a set number of tokens impacts the retrieval effectiveness. Our experimental codebase is developed using Pyserini (Lin et al., 2021) and OpenRouter API for the LLMs.

II. Related Work

A. Sparse Retrieval and BM25

BM25 (Best Matching 25) was designed by Robertson and Zaragoza (Robertson and Zaragoza, 2009). It is a probabilistic method of ranking the relevance of documents using the combination of inverse document frequency with a function of term-frequency saturation and document length normalization. Computational simplicity for use with inverted indexes as well as zero-shot performance have established it as a standard retrieval base in search engines across the spectrum, from web-scale to specialized biomedical or legal text corpora. Nogueira et al. (Nogueira et al., 2019) found that document augmentation through the generation of predicted queries to supplement each document can significantly enhance the recall of BM25 at depth 1000 without modification to the retrieval model itself.

B. LLM-Based Query Expansion

Query expansion enriches the original query with additional terms in order to bridge the lexical gap between the query and relevant documents. Traditional pseudo-relevance feedback approaches expand queries using top-retrieved documents (Carpineto and Romano, 2012), but the quality of expansion is bounded by the initial retrieval and the available vocabulary. The availability of large instruction-following language models has opened a new paradigm for *generative* query expansion.

HyDE (Hypothetical Document Embeddings) introduced by Gao *et al.* (Gao et al., 2023) is a zero-shot dense retrieval approach that generates an imaginary document from an LLM that could respond to the given query. The resulting document is encoded using a contrastive encoder model (e.g., Contriever) into an embedding representation which acts as the query for retrieving documents based on the nearest neighbor principle. By bypassing the need for relevance labels, HyDE can perform comparably to fine-tuned dense retrievers on benchmarks like TREC DL19/20 and a range of multilingual tasks. Still, HyDE does not use the generated pseudo-document as a BM25 query since it embeds the pseudo-document into an embedding vector. In case the imaginary document generated by the LLM contains terms unrelated to the context or domain of the question, HyDE mitigates some of that noise through dense filtering; however, this problem is more problematic in sparse retrieval systems.

As a solution to this problem, Query2Doc (Wang et al., 2023) leverages a few-shot prompt learning approach by supplying the LLM with query-passage examples prior to generation of the passage itself, leading to more relevant and factual expansions of queries. The generated pseudo-document consists of the generated passage concatenated with a number of repeats of the original query, which has the effect of increasing the weights of query terms in the BM25-based TF-IDF scheme. Query2Doc improves the performance of BM25 on the MS-MARCO and TREC DL datasets by 3%-15%, while being beneficial to dense retrievers both in- and out-of-domain. MuGI (Zeng et al., 2024) improves upon this by sampling multiple pseudo-documents from an LLM and using a dynamic query-weighting technique to integrate them, thus improving TREC DL BM25 by over 17%.

The latest research on Multi-Model Pseudo-Document Expansion (MPQE) (Zheng et al., 2025) introduces yet another means of improving expansion quality by using different LLMs to generate output which is then fused; according to the authors' reports, BM25 performance improves by 3–17% when applied to the MS MARCO and TREC DL datasets without any fine-tuning. This set of results clearly indicates the significance of the proposed hypothesis.

C. Rank Fusion in Information Retrieval

Reciprocal Rank Fusion (RRF), introduced by Cormack *et al.*, is a parameter-light fusion method that computes a combined score for each document as the sum of reciprocals of its ranks across constituent ranked lists. The formulation $\mathrm{RRF}(d) = \sum_r \frac{1}{k+r(d)}$, where k=60 is a smoothing constant, promotes documents that appear highly ranked in multiple lists without requiring score normalization across heterogeneous retrievers. RRF has been widely adopted in multi-stage retrieval systems and TREC submissions as a robust, zero-shot ensemble operator. Subsequent work has introduced weighted and co-occurrence-aware extensions: for instance, Exp4Fuse augments RRF with a co-occurrence factor $n(d)/10$ that rewards documents retrieved by both query routes, yielding the modified scoring $\mathrm{FR}(d) = (w_i + n(d)/10) \cdot \sum_i 1/(k + r_i(d))$.

Fusion using score-based techniques like CombSUM involves normalizing the scores obtained from information retrieval to the range $[0,1]$ and adding the values linearly. On the other hand, the α-blending technique involves linear interpolation of the values of the two retrieval paths. Previous research on hybrid information retrieval has shown that fusion based on normalized scores performs better than fusion using ranks when both the retrieval models have similar score distribution, but the latter is robust to heterogeneity in scoring scale.

D. Exp4Fuse

Exp4Fuse (Liu and Zhang, 2025), which is accepted at ACL 2025 Findings, serves as the major baseline and inspiration for our work. This paper presents a two-routes sparse retrieval approach that includes an original-query BM25 pass and a hypothesis-enhanced BM25 pass (zero-shot LLM generated query with λ repetitions) followed by modified RRF fusion. Importantly, Exp4Fuse does not encode the LLM generated text into dense embedding and does not perform cross-encoder reranking on these texts, making it relatively more efficient compared to HyDE or MuGI pipelines. Experiments on ten data sets (three from MS MARCO corpus variations and seven from BEIR subsets) indicate significant improvements over BM25, query2doc, and LameR baselines and demonstrate new SOTA results on

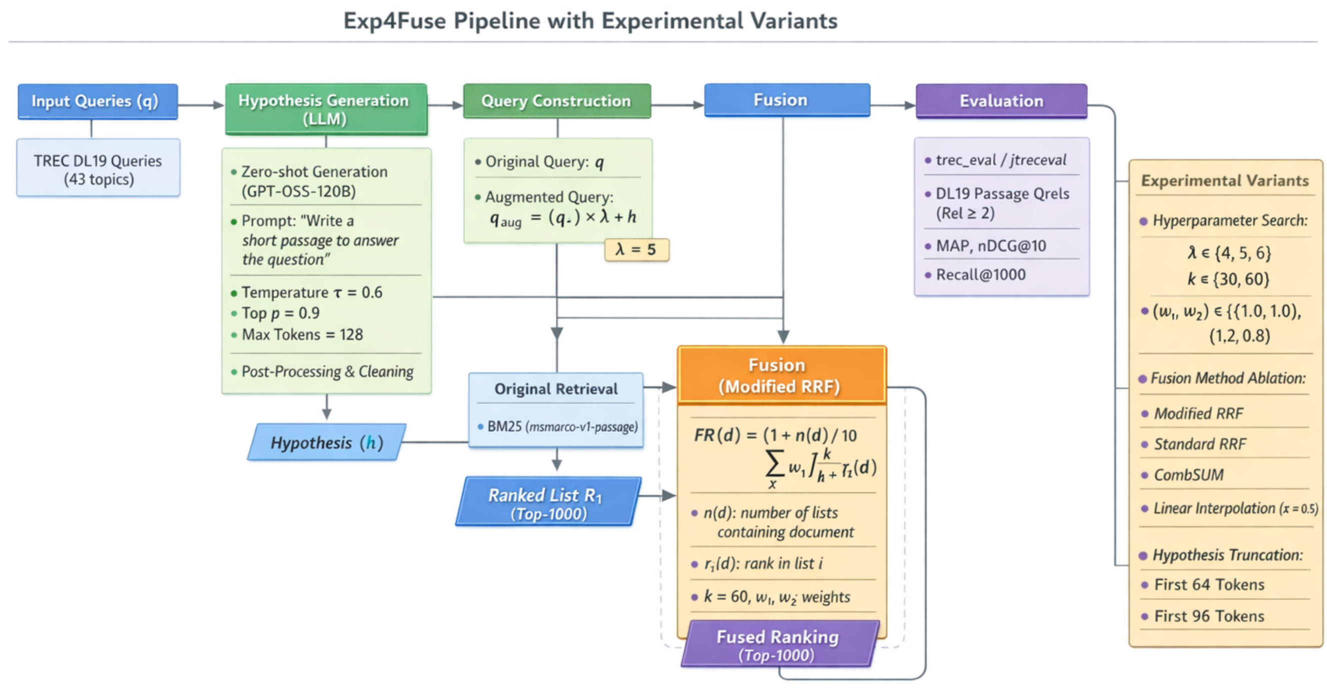

Fig. 1: Pipeline showing LLM-based query augmentation, parallel BM25 retrieval, modified RRF fusion, and evaluation with experimental variants.

several tasks when combined with learned sparse retrievers (SPLADE++).

III. Methodology

A. Pipeline Overview

The methodology is summarized in the form of a flow diagram as shown in Fig. 1.

We implement the exact pipeline described by Exp4Fuse using Pyserini's `LuceneSearcher` indexing `msmarco-v1-passage` BM25 pre-built index. For the topic set of TREC DL19 (43 queries), the Exp4Fuse pipeline consists of three steps:

Hypothesis Generation. We call the `openai/gpt-oss-120b` model through the OpenRouter API in zero-shot inference mode, using the following system prompt for each input query q: *"Please write a short passage to answer the question. Use plain text only, no markdown or formatting."* The generation is tuned according to the paper with the following parameters: temperature $\tau = 0.6$, top-$p = 0.9$, and the number of generated tokens equal to 128. We sanitize the generated hypothesis h by stripping all markdown symbols, table pipes, and whitespace characters in order to obtain the content-bearing text, which will be processed by BM25.

Retrieval. Two independent BM25 retrievals are performed for each query q: *the original route* where the raw query q is indexed, and *the hypothesis route*, where the extended query $q_{\text{aug}} = \underbrace{(q\,.)\cdots(q\,.)}_{\lambda}$

Fusion. Two ordered lists are fused into a unified list with a fusion scoring formula. Specifically, the formula Exp4Fuse used for computing the rank score for a given document d from the original RRF formula is:

$$\mathrm{FR}(d) = \left(1 + \frac{n(d)}{10}\right) \sum_{i=1}^{2} \frac{w_i}{k + r_i(d)}, \tag{1}$$

Here $n(d) \in \{1,2\}$ refers to the number of paths the document occurs in, while $r_i(d)$ refers to its ranking score for run i. The term w_i refers to the weighting parameter for each individual path and k refers to the value for the RRF smoothing factor. If the document does not appear in a particular path, then no such term is computed for it.

B. Novelty 1: Hyperparameter Grid Search

In the original Exp4Fuse study, the value for λ is fixed at 5 and that for k at 60, using equal weights for w1 and w2, i.e., w1=w2=1.0. The aim of our research is to determine if altering values for these parameters will result in any

TABLE I: Baseline Parity Results on TREC DL19 Passage Retrieval

System	MAP	nDCG@10	R@1000
BM25 (Original)	0.3013	0.5058	0.7501
BM25 (Hypothesis-Only)	0.3269	0.5475	0.7377
BM25 + Exp4Fuse	0.3208	0.5424	**0.7756**

significant performance difference for DL19. We explore the configuration parameter space with respect to the following settings:

C. Novelty 2: Fusion Method Ablation

Four fusion approaches are compared while keeping $\lambda = 5$ and $k = 60$ constant:

- **Modified RRF** (Equation 1): the Exp4Fuse baseline.
- **Standard RRF**: $\text{RRF}(d) = \sum_i 1/(k + r_i(d))$, without co-occurrence bonus.
- **CombSUM**: per-run min–max normalization of BM25 scores followed by additive combination.
- **Linear Score Interpolation**: $\alpha \cdot s_1(d) + (1 - \alpha) \cdot s_2(d)$ with $\alpha = 0.5$, applied to normalized scores.

D. Novelty 3: Hypothesis Truncation

The generated LLM passages are usually of lengths ranging between 80 to 120 tokens based on the generation setting. We investigate whether it affects the retrieval quality when we adopt shorter prefixes from the hypothesis and generate q_{aug} from 64 and 96 tokens. Other parameters remain constant as $\lambda = 5$, $k = 60$, equal weight, and modified RRF fusion.

IV. Experiments and Results

A. Experimental Setup

All experiments were performed using the TREC Deep Learning 2019 Passage Retrieval task that contains 43 queries in the MSMARCO dataset with dense relevance judgments (ranging from 0 to 3 where relevance cut-off value is 2 for `trec_eval -l 2`). The collection used in all the experiments is the entire MS MARCO Passage Corpus (around 8.8M passages). The indexing scheme employed is the pre-computed BM25 index by Pyserini with default parameters ($k_1 = 0.9$, $b = 0.4$).

B. Baseline Parity

The baseline consists of the following three retrieval runs: standalone BM25 with the original query (Original), BM25 with the hypothesis-enhanced query only (Hypothesis-Only), and the complete Exp4Fuse algorithm (Exp4Fuse). As evidenced by Table I, the pattern observed in the original paper is successfully replicated in our implementation: even without using the fusion algorithm, the hypothesis-enhanced query achieves a significant improvement in MAP and nDCG@10 over the original query.

Noteworthy is that the hypothesis-only run attains the best performance for both MAP and nDCG@10 scores but has lower Recall@1000 compared to the original run, supporting the finding reported in Exp4Fuse, wherein the expansion of a query through an LLM could result in changes in the BM25 score distribution such that some documents with higher recall and lower precision will be replaced.

TABLE II: Novelty 1: Grid Search Results (selected configurations)

λ	k	w_1	w_2	MAP	nDCG@10	R@1000
4	30	1.0	1.0	0.3330	0.5519	0.7855
4	30	1.2	0.8	0.3283	0.5502	0.7853
4	60	1.0	1.0	0.3322	0.5526	0.7855
4	60	1.2	0.8	0.3272	**0.5535**	0.7848
5	30	1.0	1.0	0.3227	0.5427	0.7756
5	30	1.2	0.8	0.3193	0.5414	0.7784
5	60	1.0	1.0	0.3208	0.5424	0.7756
5	60	1.2	0.8	0.3183	0.5438	0.7784
6	30	1.0	1.0	0.3153	0.5389	0.7690
6	30	1.2	0.8	0.3126	0.5316	0.7717
6	60	1.0	1.0	0.3140	0.5364	0.7690
6	60	1.2	0.8	0.3112	0.5314	0.7727

C. Novelty 1: Hyperparameter Sensitivity

The grid search configurations, summarized in Table II, were tried out in 12 different combinations. Several interesting facts can be highlighted. Firstly, it is clear that, when using $\lambda = 4$, the result is better than when using $\lambda = 5$ and $\lambda = 6$. The best single configuration found for the grid search was the combination of $\lambda = 4$, $k = 60$, and $w = (1.2, 0.8)$, producing nDCG@10 = 0.5535 (+0.011 compared to the default Exp4Fuse). It seems that lower repetitions make the original query more important than the augmented one and provide a clearer BM25 signal. Secondly, using the highest value $\lambda = 6$ systematically worsens all the scores, showing the negative impact of too many repetitions of the hypothesis on the relevance of the final output. Thirdly, the value of RRF constant, either $k = 30$ or $k = 60$, does not have any meaningful difference between them for all possible values of λ and weights w. Finally, using an asymmetric weighting ($w_1 = 1.2, w_2 = 0.8$), the nDCG@10 score becomes higher only if the value of λ is 4, but not 5.

D. Novelty 2: Fusion Method Ablation

The comparison between four different fusion methods, where the retrieval parameters ($\lambda = 5$, depth = 1000) are constant, is shown in Table III. The most significant observation is that CombSUM and weighted linear interpolation method ($\alpha = 0.5$) have exactly the same performance in terms of MAP, nDCG@10 and average rank, scoring 0.3309 for MAP and 0.5626 for nDCG@10, outperforming both other approaches.

The fact that the modified and normal RRF systems perform almost identically (with differences less than 0.0002 for all metrics) should also be mentioned: the additional co-occurrence term in Exp4Fuse gives no extra value for two retrieval routes on DL19. This might be due to the fact that with only two retrieval routes, the co-occurrence penalty

TABLE III: Novelty 2: Fusion Method Ablation on TREC DL19

Fusion Method	MAP	nDCG@10	R@1000
Modified RRF	0.3208	0.5424	0.7756
Standard RRF	0.3206	0.5424	0.7756
CombSUM	**0.3309**	**0.5626**	0.7765
Linear ($\alpha = 0.5$)	**0.3309**	**0.5626**	0.7765

TABLE IV: Novelty 3: Effect of Hypothesis Truncation on TREC DL19

Truncation	MAP	nDCG@10	R@1000
Full (∼100 tokens)	0.3208	0.5424	0.7756
Truncated (96 tokens)	0.3208	0.5424	0.7756
Truncated (64 tokens)	0.3208	0.5424	0.7756

reduces to an additive constant offset that doesn't affect the rankings of fused documents.

The better results obtained with score normalization compared to rank-based normalization are also expected in this particular case given the homogenous retrieval approach used: the fact that both retrieval routes are based on the same BM25 model applied to the same collection means that their scores can be considered roughly comparable, and min-max normalization should give good results.

E. Novelty 3: Hypothesis Truncation

Table IV shows the values for retrieval measures of the entire hypothesis, as well as its two reduced versions, i.e., with 64 tokens and with 96 tokens. For each of these cases, MAP is 0.3208, nDCG@10

Two possible reasons can be offered to explain this result. On the one hand, LLMs may generate such a distribution of vocabularies in their hypotheses that relevant words will naturally be grouped in the beginning. On the other hand, BM25 scoring depends on a term saturation function, so each successive occurrence of relevant terms is of lower relevance than previous ones and has less influence on the end score. An interesting direction for future research would be the study of extreme truncations (i.e., 16-32 tokens).

F. Summary and Comparison

These results are presented together in Table V, along with a comparison to the Exp4Fuse baseline. For nDCG@10 performance, the optimal solution is obtained via a linear combination of CombSUM scoring functions (0.5626). On the other hand, the highest values of MAP (0.3330) and Recall@1000 (0.7855) have been obtained with the $\lambda = 4$ parameter setting. The important point here is that these two innovations solve independent problems, which means they can be combined to yield further improvements.

G. Discussion

These findings have three practical implications for practitioners working with pipelines like Exp4Fuse. Firstly, the query repetition parameter λ proves to be extremely influential: cutting it from 5 to 4 always improves scores in our

TABLE V: Summary: Best Configuration per Novelty vs. Baseline

System	MAP	nDCG@10	R@1000
BM25 (Original)	0.3013	0.5058	0.7501
Exp4Fuse Baseline	0.3208	0.5424	0.7756
Nov. 1 Best (λ=4, k=30, equal weights)	**0.3330**	0.5519	**0.7855**
Nov. 2 Best (CombSUM / Linear)	0.3309	**0.5626**	0.7765
Nov. 3 (All truncations identical)	0.3208	0.5424	0.7756

experiments, whereas going further, up to $\lambda = 6$, deteriorates results. This follows from the fact that higher number of queries increases TF-IDF relevance of original query terms, thus diminishing impact of the vocabulary from hypotheses. Practitioners should view λ as the main hyperparameter tuning knob.

Secondly, the type of fusion function makes a noticeable difference. Score-normalization-based fusions greatly outperform rank-based fusion in the case of only BM25. Hence, practitioners should avoid using RRF as a fusion function unless there are different kinds of retrievers among the components, and this is true regardless of the extra computational costs of CombSUM and linear fusions over RRF.

Finally, hypothesis truncation does no harm at the budget sizes we have used. The practical consequence of this finding is that LLM-generated hypotheses can be made shorter without loss of retrieval effectiveness; thus, using less-expensive-to-generate hypotheses is encouraged.

V. Conclusion and Future Work

We performed a systematic analysis of the Exp4Fuse framework (Liu and Zhang, 2025), which incorporates large language models into the sparse retrieval pipeline based on the TREC DL19 dataset. Having established the baselines' equivalence, we then conducted a thorough analysis of three interesting research directions. First, our results indicate that a smaller query repetition factor ($\lambda = 4$) yields better retrieval performance than the $\lambda = 5$ value used in the paper, regardless of the value of the RRF parameter k. Second, in the case of combining homogeneous scores obtained only from the BM25 model, score normalization approaches (such as CombSUM and linear interpolation) are superior to rank-based methods (+0.020 nDCG@10). Lastly, our truncation analysis indicates that truncating hypotheses to 64 or 96 tokens does not affect retrieval performance when using modified RRF fusion.

Many possible avenues for future research exist. It might be productive to try combinations of the optimal setups in Novelties 1 and 2 (that is, $\lambda = 4$ with CombSUM fusion). Testing truncation with very small values (16-32 tokens) would help determine the true minimum length of hypotheses needed. Lastly, using these novelties on the learned sparse retrievers, like SPLADE++, would reveal whether these observations apply outside of BM25. The next logical step in the MuGI line of research would be to explore whether diverse hypotheses are beneficial for fusion in the case of Exp4Fuse.

VI. Individual Contribution

Akhilesh Negi was responsible for the implementation of the retrieval and fusion modules (`retrieval.py`, `fusion.py`), the hyperparameter grid search experiments (Novelty 1), the evaluation infrastructure (`eval_util.py`), and the experimental analysis sections of this report.

Anush Revankar was responsible for the LLM hypothesis generation pipeline (`generate_dl19_hypothesis.py`), the baseline parity replication script (`run_exp4fuse_dl19.py`), the fusion method ablation (Novelty 2), the hypothesis truncation study (Novelty 3), and the introduction, related work, and conclusion sections of this report.

Both authors together designed the experimental framework (`run_experiments.py`), analyzed results, and contributed to the final manuscript revision.

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